AlgoFlow

One page project summary

Submitted by Krish Sondagar | 15 October 2026

Application repository https://github.com/KRISH141006/AlgoFlow

Extension repository https://github.com/KRISH141006/dsa_universal_sync

Problem statement

Solving a DSA problem once does not ensure that its approach will be remembered. Accepted code, topic notes, and revision dates are often scattered across coding platforms and local files. AlgoFlow brings these records into one local workspace and schedules deliberate revision.

Objectives

Organize accepted LeetCode and GeeksforGeeks solutions; preserve exact code and problem descriptions; schedule problem and theory reviews; record improvements as solution versions; and support safe backup and restoration.

Key features

AlgoFlow uses my custom DSA Universal Sync extension to collect code and problem details from LeetCode and GeeksforGeeks. The extension can fetch solved problems together as a batch and automatically save code when a new submission is accepted. Its JSON exports can be imported into AlgoFlow. Other features include a searchable library, duplicate protection, saved solution versions, scheduled reviews, progress tracking, and backup and restore.

Technologies and tools

Python, FastAPI, SQLite, React, TypeScript, Vite, JavaScript, browser extension APIs, Pydantic, Bleach, pytest, and GitHub.

Implementation

The extension collects solutions and exports them as JSON. The React interface sends imported data and form entries to the FastAPI backend. The backend validates the data and stores problems, topics, solution versions, and review history in SQLite. Reviews follow intervals of 1, 7, 30, and 180 days, and saving a new version keeps earlier code available.

Achieved outcome

The application stores 274 imported accepted solutions with their code and descriptions, along with one existing tracker problem. All 13 automated tests passed. The project provides one place to find earlier solutions, record improvements, and manage revision. Data remains available after restarting, and backups preserve the collection. Direct extension delivery to AlgoFlow and cloud synchronization are planned improvements.